TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

Proposer	Name	Xavier Bulliard	Email	xavier.bulliard@csem.ch
Institute Name	CSEM			
Address	Rue Jaquet-Droz 1, 2002 Neuchâtel, Switzerland			

2. Technology Overview

Title of Proposed Technology	**Multifunctional encapsulation thin film for opto-electronic devices**			
Korea's 12 National Strategic Technology Fields	Semiconductors and display (1.7)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Semiconductors and displays			
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductors, display materials, parts and equipment			
Concept of Proposed Technology	Encapsulation thin film can be integrated into opto-electronic devices, such as displays or sensors, to improve performance or lifetime of the devices. We propose to functionalize thin films to generate new optical properties by adding molecular species (dyes) or nanocrystals (QDs) to the film. By controlling light energy transfer from donor-acceptor systems in the thin film, light re-emission can be improved in terms of intensity and resolution in light emitting devices for display applications. The system can be adapted to the reverse mode, for devices based on light absorption (harvesting), such as photodiode or (perovskite) solar cells, by controlling downshifting properties for instance. In this case, UV light can be cut-off without any loss in device performance, protecting the device and improving its lifetime. Both use cases will be presented in this proposal, but one will be finally selected. We propose to select one use case to demonstrate the technology, in collaboration with Korean SME. The technology is compatible with polymeric flexible thin film or glass, and the film can be further functionalized to make it hydrophobic (self-cleaning), if needed			
Rationale for Joint R&D Project	CSEM has the expertise and deposition platform for the functionalization of thin films, combining both liquid deposition (or printing) and vacuum deposition tools in clean room facilities. Materials for embedding and encapsulation, and their deposition techniques are also available. CSEM is also equipped for the characterization of thin films and coatings. However, for specific applications, such as display, CSEM is seeking the right implementation partner. A Korean SME will be ideal based on the strong expertise in the opto-electronic field in this country, and potential market. The Korean SME would dress the exact devices requirement list, characterize the devices, and participate in the upscaling and industrialization of the jointly developed materials and process. Specific materials such as QDS might also be provided by the display manufacturer.			

Key Words	1. Flexible thin films 2. Light control 3. Down-conversion; down-shifting 4. Display (or photodiodes)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3-4	End of the Project	Level 5-6
R&D Objectives	The objective is to integrate a functionalized optical flexible thin film into a display device to improve its efficiency, in terms of light re-emission. (Light harvesting system can also be considered). The following steps are considered: - Deposition of optically active species on encapsulation thin film - Characterization and demonstration of improved properties, in terms of light re-emission efficiency (or harvesting) - Integration of the functionalized film into a display device (or photodiodes/cells)			
R&D Target Specifications	- preparation of functionalized thin film – no loss in transmittance (over 95%) - characterization of optical properties of thin film: for display: light re-emission 2x as compared with a single QDs film (chosen as reference) – for light harvesting: UV cut-off below 10%, no loss in efficiency; improved lifetime by 20% - integration into display device: keep basic film performance; pass DH test (80°C / 80% RH)			
R&D Plan	**<u>Task 1: Functionalization of the film</u>** The first task is the preparation of the functionalized film. Optically active species, such as fluorescent dyes and/or quantum dots will be deposited on a plastic film, such as PET. The PET film can have barrier properties, such as metal-oxide thin films previously deposited. The deposition of active species will be done preferably by spraying or printing and cured for stabilization of the active species. The active compounds will most likely be embedded in a matrix for their proper stabilization on the surface. The composition of the matrix, polymer or hybrid organic/inorganic will be set. Preliminary stability tests against humidity/temperature will be done at that stage in climatic chamber. Basic optical performance will be measured as well, such as light absorption/emission, and the film will be transferred to Korean SME for detailed characterization. (Milestones – first fully functionalized film) **<u>Task 2. Optical characterization of the film</u>** The complete optical properties of functionalized film will be measured in this task, such as the light spectrum, luminescence efficiency, and photoluminescence quantum yield for display application. Based on the data, different iteration will be done in terms of film composition of the optically active species (dyes or QDs) and their concentration. The optimal formulation will then be			

	chosen for the next task. In case of light harvesting, In case light harvesting is preferred, light spectrum, external quantum efficiency will be measured. Wavelength cut-off and transmittance will be checked as well. Film composition will be optimized as well. (Milestone – optimized optically functionalized film, ready for integration) **Task 3. Integration into the device** In this task, the functional and characterized film will be integrated into a demonstrator display device (not necessarily final device). LED Light source (backlight) will be chosen, and other necessary optical components. The task will focus on the device initial performance characterization after assembly of all components, followed by lifetime evaluation. The plan and methodology for building a final real device will also be established. (milestone – first demonstrator of a device integrating the film)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM will focus on optical film preparation, including all necessary deposition steps. Materials selection will also be made (in discussion with the Korean SME). The material for embedding and encapsulation will also be a task of CSEM. CSEM will conduct initial optical film characterization, and aging tests in climatic chamber.
Expected R&R of Korean SME Partner	Korean SME will complete the initial optical film characterization, according to its specific device requirement. It will oversee film integration into a first prototype. CSEM will assist this step. Korean SME will also oversee device performance characterization and reliability tests. In case of light harvesting option is chosen, CSEM can assist these steps grandly, due to its expertise in cell and photodiodes (for instance based on perovskite).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint project will benefit both partners. For CSEM, it is crucial to collaborate with companies that have expertise in devices fabrication (display) and that have direct access to the market. For the Korean SME, it will benefit from new technological aspects for improved device performance and/or lifetime, and technological novelties as compared to competitors. The technological outcomes and impact are as follows: - improved light emission intensity and spectral resolution at moderate cost (for display) - ease of integration of the film (not electrically active – no need of additional power) (for display or light harvesting devices) - spectral selection and lifetime gain (for light harvesting device integrating new cells technology) As the suggested approach can be adapted to both light emitting and light harvesting devices, the companies can target two different markets through this collaboration.

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization